

Lista 8 de G.A

$$1) \pi: X = \left(-5, \frac{2}{3}, 0\right) + \lambda \left(\frac{1}{2}, 1, 1\right) \rightarrow$$

$$\begin{aligned} 3x = 2y - 16 &\rightarrow 3x - z = 0 &> 3x + 2y - 2z - 16 = 0 \\ &2y - z - 16 = 0 &\rightarrow (3, 2, -2) \end{aligned}$$

$$\text{Sen } \theta = \frac{\left| \left(\frac{1}{2}, 1, 1\right) \cdot (3, 2, -2) \right|}{\sqrt{\left(\frac{1}{2}\right)^2 + 1^2 + 1^2} \cdot \sqrt{3^2 + 2^2 + (-2)^2}} \rightarrow \text{Sen } \theta = \frac{3 \cdot \frac{1}{2} + 2 \cdot 2 - 2 \cdot 2}{\sqrt{\frac{1}{4} + 1 + 1} \cdot \sqrt{9 + 4 + 4}}$$

$$\text{Sen } \theta = \frac{3 \cdot \frac{1}{2} + 4 - 4}{\sqrt{\frac{9}{4}} \cdot \sqrt{17}} \rightarrow \text{Sen } \theta = \frac{3 \cdot \frac{1}{2}}{\frac{3}{2} \cdot \sqrt{17}} \rightarrow \text{Sen } \theta = \frac{3 \cdot \frac{1}{2}}{\frac{3 \cdot \sqrt{17}}{2}} \rightarrow \frac{1 \cdot 1}{1 \cdot \sqrt{17}}$$

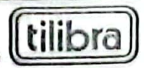
$$\text{Sen } \theta = \frac{1}{\sqrt{17/2}}$$

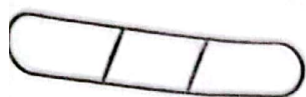
$$\text{Sen } \frac{20}{21}$$

$$\begin{aligned} b) \pi: (1, 1, 0) + \lambda(0, -1, 1) &\begin{cases} 5 - x - y + z = 2 \rightarrow x - y + z = 3 \\ z = 2 \\ x - y - 2 + 3 = 0 \rightarrow x - y - 1 = 0 \rightarrow (1, -1, 0) \\ z + 1 = 0 \end{cases} \end{aligned}$$

$$\text{Sen } \theta = \frac{|(0, -1, 1) \cdot (1, -1, 0)|}{\sqrt{0^2 + (-1)^2 + 1^2} \cdot \sqrt{1^2 + (-1)^2 + 0^2}} \rightarrow \text{Sen } \theta = \frac{|0 + 1 + 0|}{\sqrt{2} \cdot \sqrt{2}} \rightarrow \text{Sen } \theta = \frac{1}{2}$$

$$\text{Sen } \theta = 30^\circ$$





$$C) \pi: \begin{cases} x+3z=7 \\ y=0 \end{cases} \quad \begin{matrix} x=7-3z \\ y=0 \\ z=t \end{matrix} \quad \begin{matrix} x=7-3t \\ y=0 \\ z=t \end{matrix} \rightarrow (7,0,0), (1,-3,0), (1,0,1)$$

$$S: \begin{cases} x-2y-z=5 \\ y=0 \end{cases} \rightarrow \begin{matrix} x-3z=5 \\ x-3z-z=5 \end{matrix} \rightarrow (1,3,2)$$

$$\cos \theta = \frac{|(1,-3,0,1) \cdot (1,3,2)|}{\sqrt{3^2+0^2+1^2} \cdot \sqrt{1^2+3^2+2^2}} \rightarrow \cos \theta = \frac{-3+0+2}{\sqrt{9+0+1} \cdot \sqrt{1+9+4}} = \frac{1}{\sqrt{10} \cdot \sqrt{15}}$$

$$\cos \frac{1}{\sqrt{150}}$$

$$d) \pi: x = \frac{1-y}{2} = \frac{z}{3} \rightarrow X = (0,1,0) + h(1,2,3)$$

$$S: \begin{cases} 3x+y-5z=0 \\ x-2y+3z+1=0 \end{cases} \rightarrow \begin{matrix} 2x-y-2z+1=0 \\ 2x-y-2z+1=0 \end{matrix} \rightarrow (2,-1,-2)$$

$$\cos = \frac{|(1,2,3) \cdot (2,-1,-2)|}{\sqrt{(1^2+2^2+3^2)} \cdot \sqrt{(2^2+1^2+2^2)}} \rightarrow \cos = \frac{|2-2-6|}{\sqrt{1+4+9} \cdot \sqrt{4+1+4}} = \frac{4}{\sqrt{35}}$$

$$\cos = \frac{1\sqrt{50}}{10}$$

$$(2) \quad P = (0, 2, \lambda, 0) \quad Q = (1, 2, \mu)$$

$$PQ = Q - P \rightarrow (1, 2, \mu) - (0, 2, \lambda, 0) \rightarrow (1, 0, \mu, \lambda)$$

• PQ formando ângulo de 45° com π .

$$\frac{1}{\sqrt{2}} = \frac{(1, -\lambda, \mu) \cdot (0, 1, 0)}{\sqrt{1^2 - \lambda^2 + \mu^2} \cdot \sqrt{1^2}} \quad \frac{1}{\sqrt{2}} = \frac{-\lambda}{\sqrt{1 - \lambda^2 + \mu^2}}$$

• PQ formando 60° com S .

$$\frac{1}{2} = \frac{(1, -\lambda, \mu) \cdot (0, 0, 1)}{\sqrt{1^2 - \lambda^2 + \mu^2} \cdot \sqrt{1^2}} \rightarrow \frac{1}{2} = \frac{\mu}{\sqrt{1 - \lambda^2 + \mu^2}}$$

$$1 - \left(\frac{1}{\sqrt{2}}\right)^2 = \left(\frac{-\lambda}{\sqrt{1 - \lambda^2 + \mu^2}}\right)^2 \rightarrow \frac{1}{2} = \frac{\lambda^2}{1 + \lambda^2 + \mu^2}$$

$$11 \left(\frac{1}{2}\right)^2 = \left(\frac{\mu}{\sqrt{1 - \lambda^2 + \mu^2}}\right)^2 \rightarrow \frac{1}{4} = \frac{\mu^2}{1 + \lambda^2 + \mu^2}$$

$$1 - \frac{1}{2} (1 + \lambda^2 + \mu^2) = \lambda^2 \rightarrow \frac{1}{2} + \frac{1}{2} \lambda^2 + \frac{1}{2} \mu^2 = \lambda^2 \quad \times 2$$

$$11 = \frac{1}{2} (1 + \lambda^2 + \mu^2) = \mu^2 \rightarrow \frac{1}{2} + \frac{1}{2} \lambda^2 + \frac{1}{2} \mu^2 = \mu^2 \quad \times 2$$

$$11 + \lambda^2 + \mu^2 = 2\lambda^2$$

$$\mu^2 + 1 = \lambda^2$$

$$11 - 1 + \lambda^2 + \mu^2 = 4\mu^2$$

$$\lambda^2 + 1 = 3\mu^2$$

$$\mu^2 + 1 = 3\mu^2$$

$$2 = 2\mu$$

$$\mu = 1$$

$$\lambda = \pm\sqrt{2}$$

$$P = (0, 2 + \sqrt{2}, 0)$$

$$Q = (1, 2, 1)$$

$$P = (0, 2 - \sqrt{2}, 0)$$

$$Q = (1, 2, -1)$$



3) c) $X = (1, 0, 0) + \lambda(1, 1, 2)$

$\pi: x + y - z - 1 = 0 \rightarrow (1, 1, -1)$

$\cos \theta = \frac{|(1, 1, -2) \cdot (1, 1, -1)|}{\sqrt{1^2 + 1^2 + 2^2} \cdot \sqrt{1^2 + 1^2 + 1^2}} \rightarrow \cos \theta = \frac{|(1+1+2)|}{\sqrt{6} \cdot \sqrt{3}} \rightarrow \cos \theta = \frac{3}{\sqrt{18}} = \frac{3}{3\sqrt{2}}$

$\cos \theta = \frac{1}{\sqrt{2}} \quad \frac{\sqrt{2}}{2} \quad \frac{\pi}{4} \text{ radianos}$

b) $\pi: 2x - y = 0 \rightarrow (2, -1, 0)$

$\pi: -x = y = \frac{z-1}{2} \rightarrow (0, 0, 1) + \lambda(1, 1, 2)$

$\cos \theta = \frac{|(1, 1, 2) \cdot (2, -1, 0)|}{\sqrt{1^2 + 1^2 + 2^2} \cdot \sqrt{2^2 + (-1)^2 + 0^2}} \rightarrow \cos \theta = \frac{|2 - 1 + 0|}{\sqrt{6} \cdot \sqrt{5}} \rightarrow \frac{1}{\sqrt{30}}$

$\cos \theta = \frac{1}{\sqrt{30}} \quad \frac{\sqrt{30}}{30}$

a) $\pi: x = 0 \quad x = 0 \quad (0, 0, 0) + \lambda(0, 1, 1)$

$y - z = 0 \quad y = z$

$z = 1$

$\pi = (0, 0, 1)$

$\cos \theta = \frac{|(0, 1, 1) \cdot (0, 0, 1)|}{\sqrt{0^2 + 1^2 + 1^2} \cdot \sqrt{0^2 + 0^2 + 1^2}} \rightarrow \cos \theta = \frac{1}{\sqrt{2} \cdot \sqrt{1}} \quad \cos \theta = \frac{1}{\sqrt{2}}$

$\cos \theta = \frac{\sqrt{2}}{2} \rightarrow \frac{\pi}{4} \text{ en radianos}$

(4) $\pi_1: x+y+z=0 \rightarrow (1,1,1)$ Paralelo
 $\frac{\sqrt{2}}{2}$: com o plano $\pi_2: x-y=0 \rightarrow (1,-1,0)$

$$\frac{\sqrt{2}}{2} = \frac{|(a,b,c) \cdot (1,-1,0)|}{\sqrt{a^2+b^2+c^2} \cdot \sqrt{1^2+(-1)^2+0}} \rightarrow \frac{\sqrt{2}}{2} = \frac{|a-b|}{\sqrt{a^2+b^2+c^2} \cdot \sqrt{2}}$$

$$\frac{\sqrt{2}}{2} \cdot \sqrt{a^2+b^2+c^2} \cdot \sqrt{2} \rightarrow 1 \cdot \sqrt{a^2+b^2+c^2} = |a-b|$$

$$1 = |a-b| \quad a-b = \pm 1$$

(5) a) $\pi_1: 2x+y-2z-1=0 \rightarrow (2,1,-1)$
 $\pi_2: x-y+3z-10=0 \rightarrow (1,-1,3)$

$$\cos \theta = \frac{|(2,1,-1) \cdot (1,-1,3)|}{\sqrt{2^2+1^2+(-1)^2} \cdot \sqrt{1^2+(-1)^2+3^2}} \rightarrow \cos \theta = \frac{|2+(-1)+(-3)|}{\sqrt{6} \cdot \sqrt{11}}$$

$$\cos = \frac{+2}{\sqrt{66}}$$

b) $\pi_1: x = (1,0,0) + \lambda(1,0,1) + \mu(-1,0,0)$

1	0	0	$(0+\lambda+(-1), 0+0+0)$
1	0	1	$(0, -1, 0)$
-1	0	0	

$\pi_2: x+y+z=0 \rightarrow (1,1,1)$

$$\cos \theta = \frac{|(0,-1,0) \cdot (1,1,1)|}{\sqrt{0^2+(-1)^2+0^2} \cdot \sqrt{1^2+1^2+1^2}} \rightarrow \cos \theta = \frac{1}{\sqrt{1} \cdot \sqrt{3}}$$

$$\cos \theta = \frac{1}{\sqrt{3}}$$

tilibra



C) $\Pi_1: X = (0,0,0) + \lambda(1,0,0) + \mu(1,1,1)$ $\left(\cos \theta = \frac{|(0,-1,1) \cdot (0,0,-1)|}{\sqrt{1^2+1^2} \cdot \sqrt{1^2}} \right)$

λ	μ	K	$(0, \lambda, -1, K, 1)$
1	0	0	$(0, -1, 1)$
1	1	1	

$\cos = \frac{1}{\sqrt{2} \cdot \sqrt{1}}$

$\Pi_2: X = (1,0,0) + \lambda(-1,2,0) + \mu(0,1,0)$

λ	μ	K	$1, 0 + \lambda \cdot -1, 0 + K, -1$
-1	2	0	$(0, 0, -1)$
0	1	0	

$\cos \theta = \frac{1}{\sqrt{2}}$

⑥ $\Pi_1 = 2x - y + z = 0 \rightarrow (2, -1, 1)$
 $\Pi_2 = ? \quad (1, -2, 1)$

$\cos \theta = \frac{|(2, -1, 1) \cdot (1, -2, 1)|}{\sqrt{2^2+1^2+1^2} \cdot \sqrt{1^2+2^2+1^2}} \rightarrow \cos \theta = \frac{|2+2+1|}{\sqrt{6} \cdot \sqrt{6}} \quad \cos \theta = \frac{5}{6}$

$\cos \theta = \frac{5}{6}$

⑦ a) $\Pi: X-1=2Y=Z$ $X=2T+1$ $\left\{ \begin{array}{l} P_1 = (2T+1, T, 2T) \rightarrow \sqrt{(2T+1)^2 + T^2 + (2T)^2} \\ P_2 = (1, 1, 0) \\ P_3 = (2T-1, T, 2T) \rightarrow \sqrt{(2T-1)^2 + T^2 + (2T)^2} \\ P_4 = (0, 1, 1) \end{array} \right.$

$X-1=2Y \quad X=2Y+1 \quad Y=T$
 $Z=2Y \quad Z=2Y \quad Z=2T$

$\sqrt{4T^2 + T^2 + 2T+1 + 2T^2} = \sqrt{4T^2 + 2T+1 + T^2 - 2T+1 + 2T+1}$
 $9T^2 + 2T+1 = 9T^2 - 10T+3$

$12T = 2$

$T = 2/12 \quad T = 1/6$

$X = 2/6 + 1 \rightarrow 8/6 \rightarrow 2/3$

$Y = 1/6$

$Z = 2/6 \rightarrow 1/3$

$(2/3, 1/6, 1/3)$



b) $\pi = (0, 0, 1) + \lambda(4, 2, 3)$ $\left\{ \begin{array}{l} P_1 = (4\lambda, 2\lambda, 4-3\lambda) \quad \sqrt{(4\lambda-2)^2 + (2\lambda-2)^2 + (-3\lambda-1)^2} \\ P_2 = (2, 2, 5) \\ P_3 = (4\lambda, 2\lambda, 4-3\lambda) \quad \sqrt{(4\lambda)^2 + (2\lambda)^2 + (4-3\lambda)^2} \\ P_4 = (0, 0, 1) \end{array} \right.$

$x = 4\lambda$
 $y = 2\lambda$
 $z = 4-3\lambda$

$\sqrt{16\lambda^2 + 4\lambda^2 + 4 + 9\lambda^2 + 1} = \sqrt{16\lambda^2 + 4\lambda^2 + 9 + 9\lambda^2}$

$29\lambda + 9 = 29\lambda + 9 \rightarrow 0 = 0$

c) $\pi: X = (2, 3, -3) + \lambda(1, 1, 1)$ $\left\{ \begin{array}{l} P_1 = (2+\lambda, 3+\lambda, -3+\lambda) \quad \sqrt{(1+\lambda)^2 + (2+\lambda)^2 + (\lambda-3)^2} \\ P_2 = (1, 1, 0) \\ P_3 = (2+\lambda, 3+\lambda, -3+\lambda) \quad \sqrt{\lambda^2 + (1+\lambda)^2 + (1+\lambda)^2} \\ P_4 = (2, 2, 1) \end{array} \right.$

$x = 2+\lambda$
 $y = 3+\lambda$
 $z = -3+\lambda$

$\sqrt{\lambda^2 + 2\lambda + 1 + (\lambda^2 + 4\lambda + 4) + (\lambda^2 - 6\lambda + 9)} = \sqrt{3\lambda^2 + 4\lambda + 14}$

$\sqrt{3\lambda^2 + 4\lambda + 14} = \sqrt{3\lambda^2 + 4\lambda + 14}$

$12 = 4\lambda \rightarrow \lambda = 3$

$x = 5$

$y = 6$

$z = 0$

$(5, 6, 0)$

8) a) $P = (-2, 0, 1)$ e $\pi: X = (1, -2, 0) + \lambda(3, 2, 1)$ $\left\{ \begin{array}{l} d = \frac{\sqrt{0^2 + 6^2 - 12^2}}{\sqrt{13^2 + 2^2 + 1^2}} \rightarrow d = \frac{\sqrt{180}}{\sqrt{14}} \\ d = \frac{\sqrt{180}}{\sqrt{14}} \cdot \frac{\sqrt{14}}{\sqrt{14}} \rightarrow d = \frac{\sqrt{2520}}{14} \end{array} \right.$

$AB = B - A = (-2, 0, 1) - (1, -2, 0) = (-3, 2, 1)$

$AB \wedge v = \begin{vmatrix} 1 & 3 & 2 \\ -3 & 2 & 1 \\ 3 & 2 & 1 \end{vmatrix} = 1 \cdot 0 + 3 \cdot 1 - 6 + 2 \cdot 1 = -12$

$(0, 6, -12)$

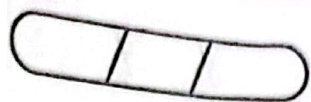
A distância é de $\frac{6\sqrt{70}}{14}$

b) $P = (1, -1, 4)$ e $\pi: X = (2, 0, -1) + \lambda(4, 3, 2)$ $\left\{ \begin{array}{l} d = \frac{\sqrt{13^2 + 22^2 + 17^2}}{\sqrt{21^2 + (-3)^2 + 2^2}} \\ d = \frac{\sqrt{169 + 484 + 289}}{\sqrt{469}} \rightarrow d = \frac{\sqrt{702}}{\sqrt{25}} \\ d = \frac{\sqrt{702}}{\sqrt{25}} \cdot \frac{\sqrt{25}}{\sqrt{25}} \rightarrow d = \frac{3\sqrt{243}}{\sqrt{25}} \end{array} \right.$

$AB = B - A = (1, -1, 4) - (2, 0, -1) \rightarrow (-1, -1, 5)$

$AB \wedge v = \begin{vmatrix} 1 & 4 & 3 \\ -1 & -1 & 5 \\ 4 & 3 & 2 \end{vmatrix} = 1 \cdot 13 + 3 \cdot 1 = 16$

$(13, 22, 7)$



$$C) P = (0, -1, 0) \text{ e } \pi: X - 2Y - 3Z = -1 \rightarrow (-3, 0, -1) + \lambda(2, 1, 1)$$

$$AB = B - A = (0, -1, 0) - (-3, 0, -1) \rightarrow (3, -1, 1)$$

$$AB \wedge v = \begin{vmatrix} 1 & 2 & 1 \\ 3 & -1 & 1 \\ 2 & 1 & 1 \end{vmatrix} \begin{matrix} 1: -2 + 2 \cdot 1 + 1 \cdot 5 \\ (-2, -1, 5) \end{matrix}$$

$$D = \frac{\sqrt{(-2)^2 + (-1)^2 + 5^2}}{\sqrt{2^2 + 1^2 + 1^2}} \rightarrow D = \frac{\sqrt{30}}{\sqrt{6}} = \frac{\sqrt{30} \cdot \sqrt{6}}{6}$$

$$A \text{ distancia } d = \frac{6\sqrt{5}}{6} = \sqrt{5}$$

$$\textcircled{9} \quad \pi: \begin{matrix} X = Y & X = Y & X = T \\ Z + 1 = Y & Y = T & Y = T \end{matrix} \rightarrow (0, 0, -1) + \lambda(1, 1, 1)$$

$$Z = Y - 1 \quad Z = T - 1$$

$$\begin{matrix} \pi_1 = X + Y = 2 & Y = 2 - X & X = T & (T, T, 2 - 2) \\ \pi_2 = X + Y + Z & X = 2 - X + Z & Y = T + 2 & \\ & -Z = 2 - 2X & Z = 2X - 2 & \end{matrix}$$

$$\rightarrow (0, 2, -2) + \lambda(1, -1, 2)$$

$$AP = P - A = (T, T, 2T - 2) - (1, 1, 0) \rightarrow (T - 1, T - 1, 2T - 2)$$

$$\sqrt{\frac{14}{3}} = \frac{\|(-1, -T + 1, 2T - 2) \wedge (1, -1, 2)\|}{\|(1, 1, 1)\|} = \frac{\|(-3 + 3, 6 - 2T - 2)\|}{\sqrt{3}} = \frac{\sqrt{12}}{3}$$

$$(-3 + 3)^2 + (6 - 2T - 2)^2 + (2T - 2)^2$$

$$-3 + 3 = 0, 3 + 3 = 6, (6 - 2T - 2)^2 = 4T^2 - 8T + 4, 2T - 2 = 2T - 2$$

$$9T^2 - 18T + 9 + 4T^2 - 8T + 4 + 4T^2 - 8T + 4$$

$$17T^2 - 34T + 17$$

tilibra

10) a) $P(-1, 3, 2)$ $\Pi: X = (1, 0, 0) + \lambda(1, 0, 0) + \mu(-1, 0, 3)$

Ponto qualquer $A = (0, 0, 0)$

X	Y	Z	
1	0	0	$(X-1).0 + Y.3 + Z.0 = 0 \rightarrow (0, 3, 0)$
-1	0	3	$-3Y = 0 \rightarrow Y = 0$

$$AB = B - A = (1, 3, 2) - (0, 0, 0) = (1, 3, 2)$$

$$(1, 3, 2) \cdot (0, 3, 0) = 9$$

$$d = \frac{9}{\|n\|} = \frac{9}{\sqrt{9}} = 3$$

b) $P(0, 0, -6)$ e $\Pi: X - 2Y - 2Z - 6 = 0 \rightarrow (1, -2, -2)$

Ponto qualquer $(2, 2, -1)$

$$AB = B - A = (0, 0, -6) - (2, 2, -1) \rightarrow (-2, -2, -2)$$

$$(-2, -2, -2) \cdot (1, -2, -2) = -2 + 4 + 4 = 6$$

$$d = \frac{6}{\sqrt{1^2 + (-2)^2 + (-2)^2}} = \frac{6}{\sqrt{9}} = \frac{6}{3} = 2$$

c) $P(1, 1, 1)$ e $\Pi: 2X - Y + 2Z - 3 = 0 \rightarrow (2, -1, 2)$

Ponto qualquer $(3, 3, 0)$

$$AB = B - A = (1, 1, 1) - (3, 3, 0) \rightarrow (-2, -2, 1)$$

$$(-2, -2, 1) \cdot (2, -1, 2) = -4 + 2 + 2 = 0$$

$$d = \frac{0}{\sqrt{2^2 + (-1)^2 + 2^2}} = 0$$

0 ponto ou seja coincido

$$\textcircled{11} \quad \begin{cases} x=2-y \\ 2-y=4+z \\ z=-2y+2 \end{cases} \rightarrow \begin{cases} x=2y-2=1 \rightarrow (1, 2, 1) \\ (2-T, T, -2T+2) \end{cases}$$

$$x=2-T$$

$$y=T \quad (2, 0, 2) + \lambda(-1, 1, 2)$$

$$z=-2T+2$$

$$\sqrt{6} = \frac{|1(2-T) + (-2)(T) + (-1)(-2T+2)|}{\sqrt{1^2 + (-2)^2 + (-1)^2}} = \frac{|-T-1|}{\sqrt{6}}$$

$$|-T-1| = 6 \quad T = -5 \text{ ou } T = -7$$

$$(2-5, 5, -2 \cdot 5 + 2) \text{ ou } (2-(-7), (-7), -2 \cdot (-7) + 2)$$

$$(-3, 5, 8) \quad (9, -7, 16)$$

$$\textcircled{12} \quad a) \quad \pi: X = (2, 1, 0) + \lambda(1, -1, 1) \quad \begin{cases} X + Y + Z = 0 \\ 2X - Y = 1 \rightarrow Y = 2X - 1 \end{cases}$$

$$(1, -1, 1) = \lambda(1, 2, -3) \quad \text{São L.I.}$$

$$AB = B - A = (0, -1, 1) - (2, 1, 0) = (-2, 2, 1)$$

$$\begin{cases} X + 2X - 1 + 2 = 0 \rightarrow Z = -3X + 1 \\ X = T \quad | \quad Y = 2T - 1 \quad | \quad Z = -3T + 1 \end{cases}$$

$$(0, -1, 1) + \lambda(1, 2, -3)$$

$$\begin{array}{c|c|c} -2 & -2 & 1 \\ 1 & 2 & -3 \\ 1 & -1 & 1 \end{array} \quad \begin{array}{c} -2 \quad -2 \\ 1 \quad 2 \\ 1 \quad -1 \end{array} \quad \begin{array}{c} -1+6-1 = (2-6-2) \\ 1-(-6) \\ 7 \end{array} \quad \begin{array}{c} \text{São reletas} \\ // \end{array}$$

$$v \wedge v \rightarrow \begin{array}{c|c|c} 1 & 2 & 3 \\ 1 & -1 & 1 \\ 1 & 2 & -3 \end{array} \quad \begin{array}{c} 1 \cdot 2 + 2 \cdot 1 + 3 \cdot 1 \\ (1, 2, 3) \end{array} \rightarrow \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$$

$$\text{Distância} = \frac{7}{\sqrt{14}} \rightarrow \frac{7\sqrt{14}}{14}$$

$$b) \pi: \frac{x+4}{3} = \frac{y}{2} = \frac{z+5}{-2} \rightarrow (2, 0, -5) + \lambda(3, 2, -2)$$

$$S: X = (2, -5, 2) + \lambda(6, -2, 1)$$

1- Verificar se são paralelos

$$(3, 2, -2) = \lambda(6, -2, 1)$$

São l.l, logo não são paralelos

2- Verificar se são reversas/concorrentes

$$AB = B - A = (6, -2, 1) - (3, 2, -2) = (3, -8, 1)$$

$$\begin{vmatrix} 3 & -8 & 1 \\ 6 & -2 & -1 \\ 3 & 2 & -2 \end{vmatrix}$$

$$(24 + 24 + 24) - (-12 - 12 + 96)$$

$$72 - 72 = 0$$

$$(12, 9, 36)$$

Logo elas são concorrentes

$$D = \frac{12^2 + 9^2 + 36^2}{\sqrt{3^2 + 2^2 + (-2)^2}}$$

$$D = \frac{144 + 81 + 1296}{\sqrt{29}}$$

$$D = \frac{1521}{\sqrt{29}}$$

$$c) \pi': \frac{x-1}{-2} = \frac{y}{2} = \frac{z}{2} \rightarrow$$

$$\frac{x-1}{-2} = \frac{y}{2} \rightarrow x-1 = -2y \rightarrow x = -2y+1$$

$$z = 2y$$

$$x = -2y+1 \rightarrow x = -2\pi+1$$

$$y = \pi \rightarrow y = \pi \rightarrow (1, 0, 0) + \lambda(-2, 1, 2)$$

$$z = 2\pi \rightarrow z = 2\pi$$

1- Verificar se são paralelos

$$\text{São paralelos} \rightarrow (2, 1, 2) = \lambda(2, 1, 2)$$

2- Verificar se são possuem ponto comum

$$O = 2\pi+1$$

$$AB = B - A = (0, 0, 2) - (1, 0, 0) \rightarrow (-1, 0, 2)$$

$$O = \pi$$

Sistema Impassível

$$z = 2\pi$$

$$\begin{vmatrix} 1 & 0 & 2 \\ -1 & 0 & 2 \\ -2 & 1 & 2 \end{vmatrix}$$

$$1 \cdot 2 + 0 \cdot 1 \cdot 6 + 2 \cdot 1$$

$$(-2, -6, -1)$$

$$D = \frac{\sqrt{(-2)^2 + (-6)^2 + (-1)^2}}{\sqrt{2^2 + 1^2 + 2^2}}$$

$$D = \frac{\sqrt{21}}{\sqrt{21}}$$

$$D = \frac{\sqrt{21} \cdot 21}{21}$$

$$D = \frac{\sqrt{81}}{21}$$

tilibra

13) a) $\pi: X = (1, 5, 7) + \lambda(3, 3, 3) + \mu(0, 1, 0)$

$(3, 3, 3) \cdot (0, 1, 1) = 4 \neq 0$

$0 + 0 + 3 = 0$ Não são paralelos

$$\left\{ \begin{array}{ccc|c} X-5 & Y-7 & Z-9 & (X-5) \cdot 0 + (Y-7) \cdot 0 + (Z-9) \cdot 1 \\ 1 & 0 & 0 & 0X + 0Y + Z - 9 = 0 \\ 0 & 1 & 0 & (0, 0, 1) \end{array} \right.$$

São transversais, com isso a distância é 0.

b) $\pi: X - Y + Z = 0$ $3X - 3$ $Z = Y - 1$ $\left\{ \begin{array}{l} X = 1 \\ Y = T \\ Z = T - 1 \end{array} \right.$ $(1, 0, -1) + \lambda(0, 1, 1)$

$2X + Y - Z = 3$ $X = 1$

$\pi: Y - Z - 1 = 0 \rightarrow (0, 1, -1)$

$(0, 1, 1) \cdot (0, 1, -1) \rightarrow 1 - 1 = 0$. São Paralelos

2 - Verificar se esta contido ou não.

$Y - Z - 1 = 0 \rightarrow 1 \cdot 0 - 1 \cdot 1 - 1 = 0$

Não esta contido

3 - Acha ponto do plano

$(0, 5, 1)$ é um ponto.

$AB = B - A \rightarrow (0, 5, -1) - (1, 0, -1) \rightarrow (-1, 5, 0)$

$D = \frac{|(-1, 5, 0) \cdot (0, 1, 1)|}{\sqrt{1^2 + 1^2}} \rightarrow D = \frac{|5|}{\sqrt{2}} \rightarrow D = \frac{5\sqrt{2}}{2}$

C) $\pi: X - Y - 1 = Z + 3$

$X = Y - 1$

$Z + 3 = Y - 1 \rightarrow Z = Y - 4$

$Y = T$

$Z = T - 4$

$X = T - 1$

$\rightarrow (-1, 0, -4) + \lambda(1, 1, 1)$

$\pi: 2x + y - 3z - 10 = 0 \quad (2, 1, -3)$

I - Verificar se é paralelo $\rightarrow \epsilon$

$(2, 1, -3) \cdot (1, 1, 1) = (2 + 1 - 3 = 0) = 0 \Rightarrow 0 = 0$

II - Verificar se está ou não contida

$2x + y - 3z - 10 = 0 \rightarrow -2 + 0 + 12 - 10 = 0$

III Como estão contidas

Distância = 0

(92) a) π_1 e π_2

$\pi_1: 2x - y + 2z + 0 = 0 \rightarrow (2, -1, 2)$

$\pi_2: 4x - 2y + 2z - 21 = 0 \rightarrow (4, -2, 2)$

$(2, -1, 2) = \lambda(4, -2, 2) \quad \lambda = 2$ faz dele L.D.

2 - Verificar se coincidem, ou não.

Pois não são coincidentes pois $0 \neq -21$

3 - Achar pontos

A $2x - y + 2z + 0 = 0 \quad (2, 2, 3) \quad 2 \cdot 2 - (-2) + 2 \cdot 3 = 0 \rightarrow 4 + 2 + 6 = 0$

B $4x - 2y + 2z = 21 \quad (2, \frac{3}{2}, 1)$

$BA = A - B = (2, 2, 3) - (2, \frac{3}{2}, 1) \rightarrow (0, \frac{1}{2}, 2)$

$n = (2, -1, 2) \rightarrow \|n\| = \sqrt{2^2 + 1 + 2} \rightarrow \sqrt{9} \rightarrow 3$

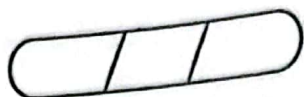
$BA \cdot n = (0, \frac{1}{2}, 2) \cdot (2, -1, 2) \rightarrow 0 - \frac{1}{2} + 4 \rightarrow 3\frac{1}{2} \rightarrow \frac{7}{2}$

$D = \frac{|-21/2|}{3}$

$D = \frac{21}{3}$

$D = \frac{21}{2} \cdot \frac{1}{3}$

$\frac{21}{6} \rightarrow \frac{7}{2}$



$$b) \pi_1: x - y + 2z - 2 = 0 \rightarrow (1, -1, 2)$$

$$\pi_2: x = (0, 0, 1) + \lambda(1, 0, 3) + \mu(-1, 1, 1)$$

$$\begin{array}{c|ccc} x & y & z & 1 \\ \hline 1 & 0 & 3 & \\ -1 & 1 & 1 & \end{array}$$

$$x \cdot 3 + y \cdot 1 + z \cdot 1$$

$$+ 3x + (-1) + 1 = 0 \rightarrow (-3, -4, 1)$$

$$(1, -1, 2) = \lambda(-3, -4, 1) \rightarrow \text{São L.I.}$$

• 2 Logo Distância = 0

$$c) \pi_1: 2x - y + z - 1 = 0 \rightarrow (2, -1, 1)$$

$$\pi_2: 4x - 2y + 2z - 9 = 0 \rightarrow (4, -2, 2)$$

$$(2, -1, 1) \neq \lambda(4, -2, 2) \rightarrow \text{São L.D.}$$

• 2 Verificar se são coincidentes

~~São coincidentes~~ Não são coincidentes $-1 \neq -9$

• 3 Achar pontos

$$A: (2x - y + z - 1 = 0) \rightarrow 2 \cdot 2 - 2 + (-1) = 0 \quad (2, 2, -1)$$

$$B: (4x - 2y + 2z - 9 = 0) \rightarrow 4 \cdot 1 - 2 \cdot (-1) + 2 \cdot 2 - 9 = 0 \quad (1, -1, 2)$$

$$BA = A - B = (2, 2, -1) - (1, -1, 2) \rightarrow (1, 3, -3)$$

$$\eta = (2, -1, 1) \rightarrow \|\eta\| = \sqrt{2^2 + (-1)^2 + 1^2} \rightarrow \sqrt{6}$$

$$BA \cdot \eta = (1, 3, -3) \cdot (2, -1, 1) \rightarrow (2 - 3 - 3) \rightarrow \frac{-4}{2} = -2$$

$$D = \frac{|-2|}{\sqrt{6}}$$

$$\rightarrow D = \frac{\sqrt{6} \cdot 2}{\sqrt{6}}$$

$$d) \pi_1 = A(0,1,6), B(5,0,1), C(2,0,0)$$

$AX \rightarrow X-A \rightarrow (X, Y, Z) - (0,1,6) \rightarrow (X-0, Y-1, Z-6)$	$X \ Y-1 \ Z-6$	$X \cdot 1 + (Y-1) \cdot 10 + (Z-6) \cdot 1$
$AB \rightarrow B-A \rightarrow (5,0,1) - (0,1,6) \rightarrow (1,0,1)$	$1 \ 0 \ 1$	$X - 10Y + 10 - Z + 6 = 0$
$AC \rightarrow C-A \rightarrow (2,0,0) - (0,1,6) \rightarrow (2,-1,-6)$	$2 \ -1 \ -6$	$X - 10Y - Z + 6 = 0$

$$(1, -10, 1)$$

$$\pi_0 = 2X + 10Y - 2Z - 16 = 0 \rightarrow (2, 10, -2)$$

$$(2, 10, -2) = \lambda(1, -10, 1) \rightarrow \text{L.D para } \lambda = 2$$

2- Ver se esta contido $-16 \neq 16$ / Não esta

3- Achar Pontos

$$B \pi_2 = 2X + 10Y - 2Z - 16 = 0 \rightarrow 2 \cdot 1 + 10 \cdot 1 - 2 \cdot 7 = 16 \rightarrow 2 + 10 - 14 = 16 \quad (1, 1, 7)$$

$$A \pi_1 = X + 10Y - Z + 16 = 0 \rightarrow -3 - 10 \cdot 1 - 3 = -16 \rightarrow -16 = 16 \quad (-3, 1, 3)$$

$$BA = A - B \rightarrow (-3, 1, 3) - (1, 1, 7) \rightarrow (-4, 0, -4)$$

$$n = (1, -10, 1) \rightarrow \|n\| \rightarrow \sqrt{1^2 + 10^2 + 1^2} \rightarrow \sqrt{102}$$

$$BA \cdot n = (-4, 0, -4) \cdot (1, -10, 1) \rightarrow -4 + 0 - 4 \rightarrow -8$$

$$D = \frac{|\sqrt{102}|}{-8} \rightarrow D = \frac{-\sqrt{102}}{8}$$

$$\begin{aligned}
 (15) \quad \pi: X+Z=5 & \quad X=T & \quad \left(\begin{aligned} \vec{S}: X &= (2, 1, 1) + \lambda(4, 2, -3) \\ Y+Z &= 5 & Y=1 & (0, 5) + T(1, 0, -1) \\ Z &= 5-T \end{aligned} \right)
 \end{aligned}$$

$$\begin{array}{c|ccc|l}
 X & Y & Z & S & \\
 \hline
 2 & 1 & 1 & 5 & (X) \cdot (-2) + (Y) \cdot (-1) + (Z) \cdot (-2) \\
 4 & 2 & -3 & & -2X + Y - 2Z + 10 \\
 1 & 0 & -1 & & -2X + Y - 2Z + 9 = 0
 \end{array}$$

$$\frac{|-2x_0 + y_0 - 2z_0 + 9|}{\sqrt{(-2)^2 + 1^2 + (-2)^2}} = \frac{-2x_0 + y_0 - 2z_0 + 9}{3} = 2$$

$$|-2x_0 + y_0 - 2z_0 + 9| = 6 \rightarrow -2x_0 + y_0 - 2z_0 + 9 = \pm 6 - 9$$

$$\pi_1: -2x + y - 2z = -3$$

$$\pi_2: -2x + y - 2z = +6$$